AXION BIOSYSTEMS

DATA ACQUISITION QUICK-START

INTRODUCTION

The Maestro Pro and Maestro Edge are powerful, yet simple, platforms for measuring and quantifying electrical activity from cells and networks. With one button setup, simply insert a multiwell MEA plate into the MEA chamber and push the button, which will dock the plate, set environmental controls, and configure the software so that you can focus on the science. Below are more detailed, step-by-step instructions for recording raw data from the Maestro Pro or Maestro Edge. For more information, consult the AxIS Navigator User Guide.

DATA ACQUISITION IN AXIS NAVIGATOR

1. Turn on the Maestro using the switch on the left side.
2. Open *AxIS Navigator*.

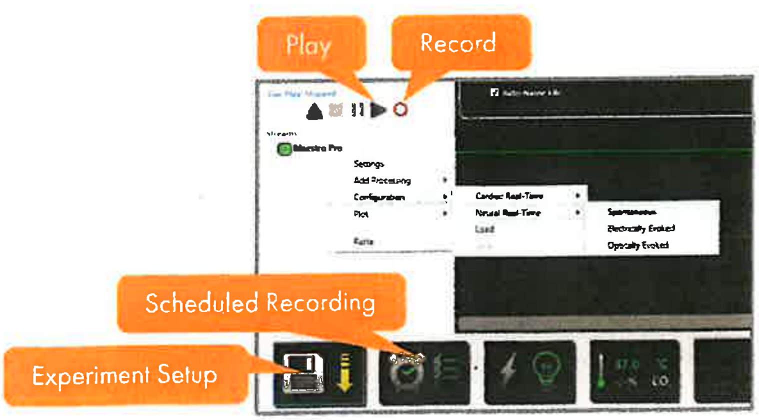
   Open

   Note: *The data stream will say "Maestro Pro" or "Maestro Edge" when connected and "No Device Connected" when it is not or while initiating.*
3. Allow the temperature to stabilize as indicated by the Environmental Control thumbnail.

   Note: *The **Maestro Pro** also displays temperature and CO_2 status on the touch screen.*
4. Right-click on the **Maestro Pro** or **Edge** stream and select **Configuration** to apply the desired configuration as outlined below.

   Configure

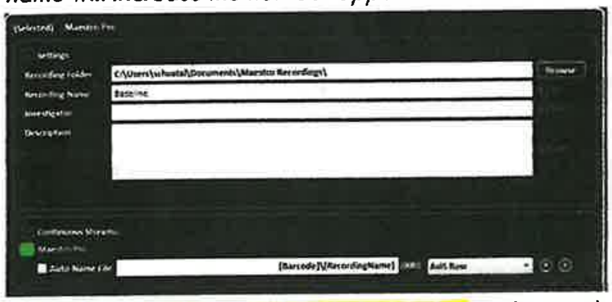
   a. Cardiac Real-Time → Field Potentials, LEAP, or Contractility with the option of Paced if needed.
   b. Neural Real-Time → Spontaneous, Electrically Evoked, or Optically Evoked.



5. Click the **Experiment Setup Properties** module.

   File Path
6. Enter a name for the recording in the **Recording Name** field. Optional: Enter the **Investigator** and **Description** in the respective fields

7. Optional: Uncheck **Auto Name File** below the **Maestro** menu to enter a custom file name. By default, the file name is the **Recording Name** specified above, and the file is saved to a folder corresponding to the plate barcode. Auto-naming macros are available to add descriptive information to folder and file names. ***Note:*** *File names are appended with a number starting with 000. Recording another file with the same name will increase the number appended to the end.*



Insert plate

8. Place the MEA plate into the MEA chamber and press the button on the Maestro to engage the MEA and close the door. The Maestro will automatically begin controlling CO_2.
9. Allow the MEA to equilibrate while docked in the Maestro for 10-20 minutes prior to recording.

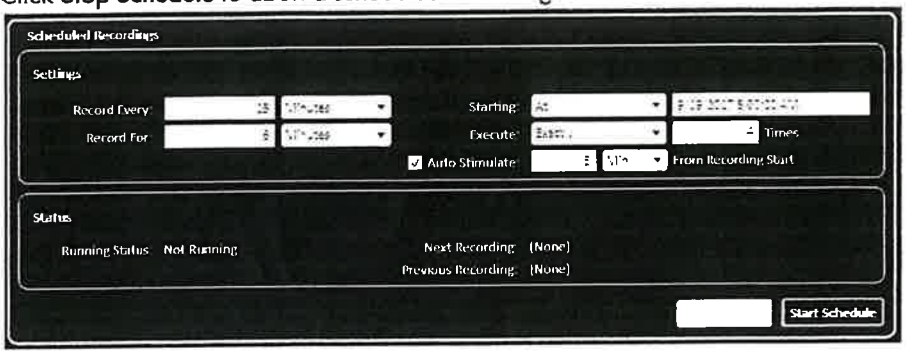
Config Plate map

10. Double click on the **Active Plate** figure to enter a plate map. ***Note:*** *Plate maps may be imported from a .platemap file or a previously recorded .raw file using the* ***Import*** *button.*

- Play
- offset
- Record or Schedule Record

11. Click **Play** (black triangle) to view the raw data and begin the voltage offset, indicated on the grey status bar. The voltage offset runs each time *AxIS Navigator* begins measuring data from a stopped state.
12. Wait for the voltage offset to complete (~30 sec in neural modes and ~1 min in cardiac modes).
13. To record manually:
    13.1. Click the **Record** button (red circle) to begin recording.
    13.2. Click the **Record** or **Stop** button to stop recording.
14. To record a preset duration:
    14.1. Click on the **Scheduled Recording Setup** module.
    14.2. Enter the desired file duration in the **Record For** field.
    14.3. Enter when to begin recording in the **Starting** field.
    14.4. Enter how many times to create a recording in the **Execute** field. For a single recording, select **Once.**
    14.5. If creating more than one recording from a file, enter the time between the start of each recording in the **Record Every** field.
    14.6. Click **Start Schedule** to begin recording.

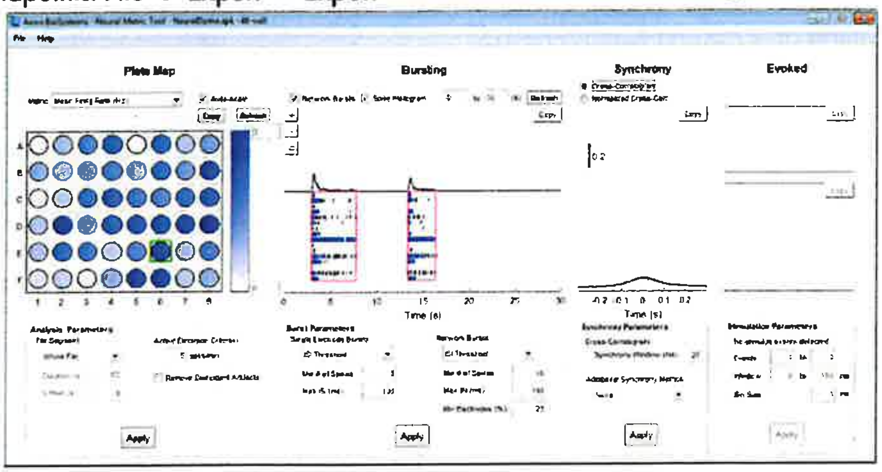
14.7. Click **Stop Schedule** to abort a scheduled recording.



AXION
BIOSYSTEMS

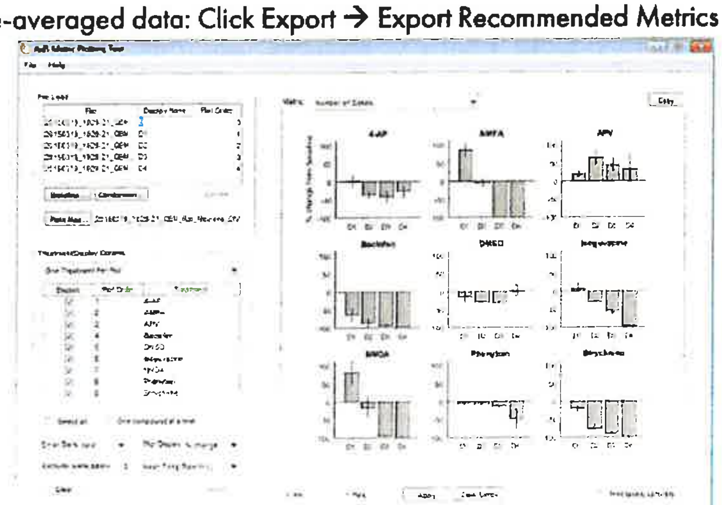
NEURAL METRIC TOOL

1. Load Data: Click File → Load Axion Spike File ....
2. Adjust Analysis, Burst, Synchrony, and Stimulation Parameters if needed.
3. Select a well in the Plate Map to display Bursting, Synchrony, and Evoked plots for that well.
4. To export figures: Click Copy.
5. To export endpoints: File → Export → Export Recommended Metrics to CSV.



AXIS METRIC PLOTTING TOOL

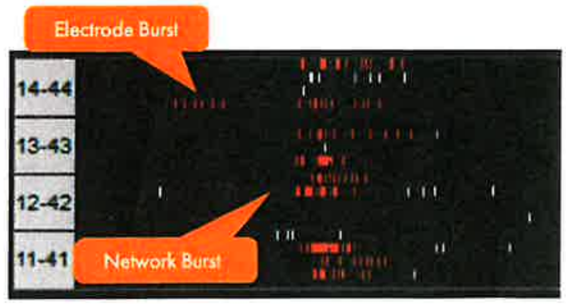
1. Load baseline data: Click Baseline. Select the baseline .csv file.
   Note: The plate map used for analysis is read from the baseline file. To use a different plate map, click Plate Map and select a .platemap file or statistics compiler .csv file containing a platemap.
2. Load comparison data: Click Comparison. Select one or more .csv files from the same folder for comparison. The Baseline and Comparison files must all have the same plate type.
3. Select metric of interest using the Metric drop-down menu.
4. Adjust Treatment/Display options if needed.
5. To export figures: Click Copy.
6. To export replicate-averaged data: Click Export → Export Recommended Metrics to CSV.



NEURAL ANALYSIS QUICK-START

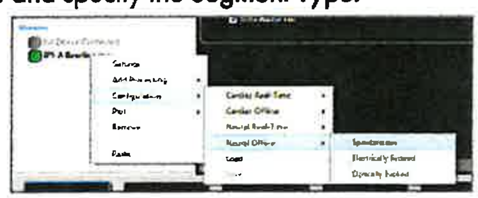
INTRODUCTION

Neural cultures form intricate networks that produce complex patterns of activity. *AxIS Navigator* computes a variety of endpoints to analyze and quantify neural network activity. Adjustable parameters automate the detection of neuronal action potentials, individual electrode bursting, network bursting, and synchrony. The following guide outlines basic instructions for analyzing neural data in *AxIS Navigator*, the *Neural Metric Tool*, and the *AxIS Metric Plotting Tool*. For more information, consult the AxIS Navigator User Guide.



NEURAL ANALYSIS IN AXIS NAVIGATOR

1. Load file(s) to analyze:
   1.1. Single file: Click **File → Open Recording**. Multiple files: Click **File → New Batch Process**.
   1.2. Define the plate map, if not already defined. Double-click on the **Active Plate** in the top left.
   Note: Define plate maps prior to running a batch process by opening and updating each file individually.
2. Set the analysis window. Double-click on the **File/Batch Process** and specify the **Segment Type**.
3. Apply the neural analysis configuration: Right-click on **File/Batch Process** in the Streams window and select **Configuration → Neural Offline → Spontaneous**.
   Note: If stimulating, select Electrically Evoked or Optically Evoked.



4. Specify file(s) to save in the **Experiment Setup Properties** module: Select **Advanced Metrics** in the **Statistics Compiler** menu to generate the results spreadsheet. Select **AxIS Spike** in the **Spike Detector** menu to save a .spk file if using the *Neural Metric Tool*.
5. Optional: If analyzing a single file, uncheck **Auto-Name File** to specify a file name for the output .csv file. By default, the name will be based on the source file name. Auto-naming macros are available to add descriptive information to file names.
6. Click **Play** to view the data.
7. Optional: Double-click on the **Spike Detector** or **Burst Detector** in the Streams window to adjust the settings if needed.
8. If playing, click **Stop** to stop the playback.

9. Record analysis file(s). Single File: Click the **Record** button (red circle). Batch Process: Click **Start Batch Process**.